

LoRA-Adapted Foundation Models vs. Specialist U-Net for Cross-Dataset Polyp Segmentation

Project Report, Research Method in Computing

MSU Summer 2026, Master's Final Project

Mikko Palis

1. Introduction

I wanted to answer one question with this project: can the Segment Anything Model (SAM), adapted with Low-Rank Adaptation (LoRA) while its backbone stays frozen, match a specialist U-Net on the standard PraNet benchmark, and lose less accuracy on datasets neither model trained on?

The question matters because clinical deployment is a cross-dataset problem. A segmentation model trained at one clinic meets different endoscope hardware, lighting, and patient populations at the next clinic. Accuracy on the training benchmark does not measure this. The seen-to-unseen drop does, so that drop is the number I built the study around.

I trained six model variants under one shared protocol, three random seeds each, on one NVIDIA A100 GPU, and evaluated four untrained oracle-box baselines, for 22 runs in total. Four findings came out of it:

1. **SAM-ViT-H + LoRA scores highest on the unseen splits.** It reaches 0.806 mean Dice on the three unseen datasets against U-Net's 0.755, and its seen-to-unseen drop (0.082) is about half of U-Net's (0.145). It trains 830,177 parameters, about 3% of the 24.4M that the ImageNet-initialized U-Net trains end-to-end.
2. **The specialist still wins at home.** U-Net posts the best seen-data score (0.900 mean Dice) and trains twelve times faster.
3. **A measured diagnosis of MedSAM's deficit.** MedSAM + LoRA landed last (0.661 unseen). A controlled three-arm experiment split its 0.100 deficit against SAM-ViT-B: a preprocessing bug in my pipeline cost 0.069 Dice; with backbone and preprocessing matched, the MedSAM checkpoint still trails generic SAM by 0.036. A separate 0.045 backbone-size effect, measured on generic SAM, is comparable to that residual. My first reading was that medical pretraining is a net drag. I retracted that after the controlled runs.
4. **Prompt dependence quantified.** With its two contracts honored (a box prompt and min-max inputs), untrained MedSAM posts the highest number in the oracle tier (0.925 unseen), an upper bound rather than a peer of the trained rows. A ground-truth box lifts SAM-ViT-H by +0.099 but lifts corrected MedSAM by +0.200. MedSAM is a specialized model, and the pattern is consistent with an encoder that relies on the box for localization and pays when the box is taken away.

The contributions are a controlled comparison of three model families (six trained variants) under one protocol, two matched pairings that separate the pretraining source from the backbone size, a measured preprocessing effect, a documented root-cause investigation of a preprocessing confound, and a reproducible config-driven pipeline with all results consolidated in one summary.

2. Literature Review

Specialist polyp segmentation networks. U-Net [1] established the encoder-decoder architecture with skip connections that dominates medical image segmentation. PraNet [2] defined the evaluation protocol this study adopts: train on 1,450 images from Kvasir-SEG and CVC-ClinicDB, test on held-out portions of the same two datasets (seen) and on three datasets the model never saw (CVC-ColonDB, ETIS-LaribPolypDB, CVC-300). The split sizes come from the PraNet data release [15]; the paper itself describes an 80/10/10 split. PraNet's tables show its specialist baselines lose accuracy on the unseen splits (U-Net falls from 0.818 on Kvasir to 0.512 on ColonDB), which made cross-dataset generalization a standard metric for this task. Polyp-PVT [3] pushed specialist accuracy further with a pyramid vision transformer backbone. Both are strong on the benchmark; both are trained end-to-end on a few thousand polyp images, which bounds how much visual diversity they can absorb.

Segmentation foundation models. SAM [4] trained a ViT image encoder and a promptable mask decoder on SA-1B, a corpus of 11M images and 1.1B masks. Zero-shot SAM segments natural images well but underperforms on medical imagery, where object boundaries follow tissue contrast rather than natural-image edges. MedSAM [5] fine-tuned SAM ViT-B on about 1.57M medical image-mask pairs across ten modalities and reported that it "ranked in first place on most tasks, surpassing the performance of the U-Net and DeepLabV3+ specialist models" [5], endoscopy among them, with one structural condition: every training step supplied a bounding-box prompt, and the specialist baselines in that comparison received the box as well. The model was never asked to localize an object on its own.

Parameter-efficient fine-tuning. LoRA [6] freezes a pretrained weight matrix and learns a low-rank update, which cuts trainable parameters by orders of magnitude while it keeps the pretrained representation intact. SAMed [7] applied LoRA to a frozen SAM encoder for medical segmentation and is the method this project adapts: LoRA on the attention projections, no prompts at inference. SAMed keeps and fine-tunes SAM's own mask decoder; this project replaces that decoder with a small CNN decoder, which is the one departure from the SAMed recipe.

The question I did not find answered. The surveyed work reports benchmark accuracy. None of it says whether LoRA adaptation of a foundation model buys *generalization*, measured on the same seen/unseen protocol the specialists were built on, under one training protocol, with the cost of each option recorded. Two published claims also sit in tension: MedSAM reports beating box-prompted specialist baselines on endoscopy, and the PraNet protocol is exactly the setting

where that claim can be tested without a prompt oracle on either side. I did not find a published run of that test. This study runs it.

3. Problem Formulation

Research problem. Specialist polyp networks fit their training distribution tightly and lose accuracy on data from other clinics. Foundation models carry broad visual priors but are expensive to fine-tune fully and, zero-shot, underperform on medical imagery. The problem is to find out whether parameter-efficient adaptation keeps the foundation model's generalization at a cost a specialist would accept, and to trace any difference to its actual cause rather than to a confound.

Research questions.

- **RQ1.** Does LoRA-adapted SAM match a specialist U-Net on the seen PraNet test splits?
- **RQ2.** Does it degrade less on the three unseen splits, and by how much?
- **RQ3.** What does each option cost in trainable parameters, checkpoint size, and training time on identical hardware?
- **RQ4.** Does medical pretraining (MedSAM) help or hurt under prompt-free adaptation, once backbone size and preprocessing are controlled?
- **RQ5.** How much of each model's deficit is prompt dependence, measured against an oracle-box upper bound?

Hypotheses. H-main: the frozen SAM backbone's broad pretraining transfers, so the LoRA variant shows a smaller seen-to-unseen drop than U-Net. H-medical: medical pretraining gives MedSAM an advantage over generic SAM at equal backbone size. The data did not support the second hypothesis. Section 6 reports the measured comparisons, including the part of the deficit my own pipeline caused.

Significance. If a frozen foundation backbone plus a sub-1M-parameter adapter generalizes better than a specialist, then a site with modest data and compute has a measured reason to test adapters against end-to-end training of a pretrained specialist on its own data. The MedSAM analysis also shows how a preprocessing contract violation can look like a scientific finding. That lesson applies to any pipeline that mixes checkpoints with different input contracts.

4. Methodology

4.1 Datasets and protocol

The study follows the PraNet five-split protocol: the same five datasets and the same split sizes as the PraNet data release [15], as configured in `configs/base.yaml`.

Dataset	Role	Images
---------	------	--------

Kvasir-SEG [8]	900 train / 100 seen test	1,000
CVC-ClinicDB [9]	550 train / 62 seen test	612
CVC-ColonDB [10]	unseen test	380
ETIS-LaribPolypDB [11]	unseen test	196
CVC-300 (EndoScene) [12]	unseen test	60

Training uses 1,450 images. The two seen test splits measure learning ability; the three unseen splits measure generalization. The primary derived metric is the seen-to-unseen drop: mean seen mDice minus mean unseen mDice (the Gap column in the tables).

4.2 Models

Model	Backbone	Trainable params	Role
U-Net	ResNet-34 encoder (ImageNet init)	24,436,369	specialist baseline
SAM-ViT-H + LoRA	frozen SAM ViT-H	830,177	main contribution
SAM-ViT-B + LoRA	frozen SAM ViT-B	322,273	backbone-size control
MedSAM-ViT-B + LoRA	frozen MedSAM ViT-B	322,273	pretraining-source arm
MedSAM min-max / ctrl	frozen MedSAM ViT-B	322,273	normalization experiment arms
Vanilla SAM / MedSAM (4 variants)	frozen, untrained	0	oracle-box upper bound

The LoRA models adapt the SAMed recipe: the ViT encoder is frozen, rank-4 LoRA (alpha 8, dropout 0.1) wraps the merged query/key/value projection of every encoder attention block (one shared low-rank update across Q, K, and V, since SAM's encoder stores them as one linear layer), and a small CNN decoder reads a mask from the encoder features with no prompt at inference. SAMed targets Q and V separately and keeps SAM's mask decoder; both differences are noted in Section 7. SAM-ViT-B + LoRA shares MedSAM's backbone size but keeps generic SAM weights, so the pair separates backbone capacity from pretraining source. Two caveats bound these pairings, and I want them stated before the results rather than after. The LoRA parameter count scales with the backbone (830K on ViT-H against 322K on ViT-B), so the size comparison moves adapter capacity together with backbone capacity. And MedSAM's starting weights bundle two changes its authors made together, medical training data and a box prompt supplied on every training example, so the pretraining pair measures the checkpoint as delivered, not the medical data alone.

4.3 Training protocol

All trained models share one protocol: 352×352 inputs, batch size 4, AdamW [18] (learning rate 5e-5, weight decay 1e-4), cosine schedule, combined Dice + BCE loss, up to 50 epochs with early stopping at patience 8, checkpoint selection by best validation Dice, three seeds (42, 43, 44), one NVIDIA A100-SXM4-40GB. Augmentation: flips, rotation, and color jitter via Albumentations

[20]. Input normalization is bound to the model key: ImageNet standardization for SAM and U-Net (the same mean and std that SAM's native `preprocess` applies), per-image min-max [0,1] for the corrected MedSAM arms (the contract in MedSAM's released inference script [16]).

4.4 Evaluation

All five splits are scored with the PraNet metric suite: mDice and mIoU (primary), weighted F-beta [17], S-measure [13], E-measure [14], and MAE. Reported numbers are means with standard deviation over the three seeds. Cost metrics (trainable parameters, checkpoint size, wall-clock training time, device name) are recorded in each run's `metrics.json`.

4.5 Oracle-box baselines

Four untrained baselines run inference only, prompted with a bounding box derived from the ground-truth mask (5 px padding): vanilla SAM ViT-H, vanilla SAM ViT-B, vanilla MedSAM under ImageNet normalization, and vanilla MedSAM under its own min-max normalization. The ground-truth box reveals the polyp's location, so these numbers form an upper bound. I keep them in a separate tier and never rank them against the trained models. The box protocol is standard in the medical-SAM literature, and I confirmed it with my advisor before the final runs (`docs/PROJECT_PLAN.md`).

4.6 The MedSAM three-arm experiment

A review on 2026-07-28 found that my pipeline fed every model ImageNet-standardized inputs. That is correct for SAM by construction and wrong for MedSAM, whose frozen encoder was fine-tuned on per-image min-max [0,1] inputs. The obvious fix was a re-run under min-max. I did not take it, because it would have changed two things at once: min-max normalization is exactly invariant to brightness and contrast jitter, so correcting the normalization also weakens the effective augmentation. A one-arm fix would have left me unable to say which change moved the number. The design therefore uses three arms:

Arm	Normalization	Brightness/contrast jitter	Role
medsam	ImageNet	on	published baseline, never re-run
medsam_ctrl	ImageNet	off	augmentation control
medsam_minmax	min-max	on (mostly cancelled by min-max)	corrected arm

The controlled normalization effect is `medsam_ctrl` $\rightarrow$ `medsam_minmax`; the size of the augmentation confound is `medsam` $\rightarrow$ `medsam_ctrl`.

5. Implementation Details

5.1 Architecture of the codebase

The pipeline is config-driven. `configs/base.yaml` holds the defaults and the per-model blocks; `configs/run.yaml` holds the values the study actually used (batch size, epochs, learning rate, patience) and selects the model, seed, and output paths. `train.py` trains any model key under the identical protocol; `evaluate.py` scores all five splits; `zeroshot_eval.py` runs the untrained oracle-box baselines; `aggregate_results.py` consolidates every run's `metrics.json` into `results/summary/` without a GPU. Model construction lives in `src/models/` (U-Net via `segmentation-models-pytorch` [19]; `sam_adapter.py` for the LoRA wrapper and light decoder; `zeroshot.py` for the oracle baselines). Metrics live in `src/metrics/segmentation.py`. Training ran on Google Colab A100 instances through `notebooks/train_colab.ipynb`, which mirrors checkpoints and results to Google Drive and skips already-trained pairs.

Two implementation decisions matter more than the rest. First, the training protocol (normalization and photometric augmentation) is bound to the model key in `src/config.MODEL_SPECS` and cannot be set in YAML. I chose this after the normalization bug: a protocol mismatch now requires a code change with review instead of a quiet config edit. Second, checkpoints follow one path contract (`checkpoints/<model>/seed<seed>/best.pt`), which lets the aggregator and the benchmark notebook find every run without hand-listing them.

5.2 Challenges and how they were addressed

The MedSAM normalization bug. The largest challenge was diagnostic. MedSAM + LoRA landed last on every split (0.661 unseen, 0.500 on ETIS-LaribPolypDB), and my first write-up blamed medical pretraining. That was the wrong call, and the honest reason is that I had not checked the input contract. A root-cause investigation (`docs/MEDSAM_INVESTIGATION.md`) ranked four hypotheses and found that the pipeline's single global ImageNet normalization violated MedSAM's input contract: its encoder was fit on $[0,1]$ inputs and received inputs in roughly $[-2.1, +2.6]$. The fix bound normalization to the model key. The confound analysis then showed the naive fix would itself be confounded (min-max absorbs brightness/contrast jitter), which produced the three-arm design in Section 4.6. I kept the published `medsam` arm and never re-ran it; the corrected arms got separate model keys, checkpoints, and results paths, so the retraction and the correction both stay visible in the results.

Prompt protocol for the untrained tier. A zero-shot SAM needs a prompt. A ground-truth box is an oracle: it hands the model the location. The tradeoff was between dropping the untrained tier, which loses the upper bound, and reporting it next to trained models, which invites a false comparison. I took a third route: keep the oracle tier in a separate table, flag the negative seen-

to-unseen gap as the sign that these numbers track split difficulty rather than learning, and get my advisor's sign-off on the box protocol before the final runs.

Compute discipline. Colab sessions are preemptible and quota-bound. The trainer restores finished runs from Drive and skips them, so a lost session costs only the run in flight. Early stopping kept several 50-epoch budgets to 28 to 49 actual epochs.

Testing without a GPU. Prompt-derivation math and the config/aggregation layer are torch-free and covered by 70 unit tests (`pytest tests/`), so I could check the logic that defines the experiment on my laptop before spending GPU time.

6. Results and Analysis

6.1 Accuracy and cost (trained, prompt-free tier)

Means $\pm$ standard deviation over seeds 42, 43, 44; one A100.

Model	Seen mDice	Unseen mDice	Gap	Trainable params	Ckpt MB	Train min
SAM-ViT-H + LoRA	0.887 $\pm$ 0.008	0.806 $\pm$ 0.011	0.082 $\pm$ 0.004	830,177	2,533	120
SAM-ViT-B + LoRA	0.865 $\pm$ 0.005	0.761 $\pm$ 0.005	0.104 $\pm$ 0.007	322,273	349	29
U-Net (ResNet-34)	0.900 $\pm$ 0.008	0.755 $\pm$ 0.014	0.145 $\pm$ 0.014	24,436,369	98	10
MedSAM + LoRA (min-max norm)	0.834 $\pm$ 0.001	0.725 $\pm$ 0.004	0.110 $\pm$ 0.004	322,273	349	30
MedSAM + LoRA (ImageNet norm, as published)	0.819 $\pm$ 0.006	0.661 $\pm$ 0.008	0.158 $\pm$ 0.011	322,273	349	31
MedSAM + LoRA (ImageNet norm, jitter control)	0.815 $\pm$ 0.001	0.655 $\pm$ 0.002	0.160 $\pm$ 0.002	322,273	349	30

RQ1 (seen). U-Net wins at home: 0.900 against SAM-ViT-H's 0.887. The specialist fits the training distribution tightest, including the best ClinicDB score of any trained model (0.897).

RQ2 (unseen). SAM-ViT-H + LoRA leads every unseen split and holds 0.806 mean unseen Dice against U-Net's 0.755. Its seen-to-unseen drop is 0.082, about half of U-Net's 0.145. SAM-ViT-B + LoRA (0.761 unseen) lands level with U-Net (0.755) on the unseen mean, inside the seed spread

(U-Net's seed 44 reaches 0.774), while it trains 1.3% of U-Net's parameters; its smaller drop (0.104 against 0.145) is the separated number. The separation is widest on ETIS-LaribPolypDB, the hardest unseen split: SAM-ViT-H 0.745, SAM-ViT-B 0.692, U-Net 0.690.

RQ3 (cost). U-Net trains fastest (10 min) with the smallest checkpoint (98 MB). SAM-ViT-H + LoRA costs 120 min and a 2.5 GB checkpoint, dominated by the frozen backbone. SAM-ViT-B + LoRA is the efficiency pick: 29 min, 349 MB, and a smaller seen-to-unseen drop than U-Net. Accuracy per trainable parameter favors the adapters; accuracy per minute favors U-Net. Which one matters depends on whether the target is the seen benchmark or the unseen splits. The whole study (18 trained runs) took about 750 GPU-minutes, or 12.5 A100-hours. At Colab rates that is roughly 15 dollars of compute. The GPU-hours are measured; the dollar figure is an estimate from the published rate.

6.2 Per-split behavior (mean over 3 seeds)

Model	Kvasir (seen)	ClinicDB (seen)	ColonDB	ETIS	CVC-300
SAM-ViT-H + LoRA	0.906	0.869	0.785	0.745	0.887
SAM-ViT-B + LoRA	0.899	0.831	0.731	0.692	0.859
U-Net (ResNet-34)	0.903	0.897	0.729	0.690	0.847
MedSAM + LoRA (min-max)	0.883	0.785	0.732	0.584	0.859
MedSAM + LoRA (as published)	0.852	0.786	0.656	0.500	0.827
MedSAM + LoRA (jitter control)	0.853	0.777	0.650	0.497	0.819

The top three models sit within 0.035 of each other on the seen splits and spread by 0.051 on the unseen ones; the MedSAM arms trail on both.

6.3 The MedSAM diagnosis: matched pairings (RQ4)

Four controlled comparisons, each changing one factor:

Comparison	Factor changed	Unseen mDice effect
SAM-ViT-H → SAM-ViT-B	backbone shrinks (SAM weights fixed)	0.806 → 0.761 (-0.045)
medsam → medsam_ctrl	brightness/contrast jitter off (ImageNet norm fixed)	0.661 → 0.655 (-0.006)
medsam_ctrl → medsam_minmax	normalization corrected (jitter-controlled)	0.655 → 0.725 (+0.069)

SAM-ViT-B → MedSAM-minmax	pretraining source (backbone and preprocessing matched)	0.761 → 0.725 (-0.036)
---------------------------	---	---------------------------------

Of MedSAM's original 0.100 unseen deficit against SAM-ViT-B, 0.069 was my preprocessing bug and 0.036 is the residual effect of the MedSAM checkpoint at matched size. That residual and the backbone effect (0.045) are close, and the seed spreads behind them run from 0.004 (the ViT-B arms) to 0.011 (SAM-ViT-H), so the two read as comparable in size rather than strictly ordered. The three effects sum to the deficit only with the jitter term included ($-0.006 + 0.069 + 0.036 = 0.100$). The checkpoint effect itself bundles MedSAM's medical training data with its box-prompted training objective; Section 6.4 points to the objective as the likely driver. The jitter confound was small (0.006), which shows the augmentation interaction did not drive the result. On this evidence I retracted the earlier claim that medical pretraining is a net drag and accounts for most of the deficit.

6.4 Oracle-box upper bound (RQ5)

Untrained, prompted with a ground-truth-derived box; a separate tier, not a fair peer.

Model	Seen mDice	Unseen mDice	Gap
MedSAM ViT-B (oracle-box, min-max norm)	0.912	0.925	-0.012
SAM ViT-H (oracle-box)	0.860	0.905	-0.046
SAM ViT-B (oracle-box)	0.821	0.880	-0.060
MedSAM ViT-B (oracle-box, ImageNet norm)	0.800	0.845	-0.045

I read two things from this table. First, the normalization bug replicates with zero training: the same MedSAM weights score 0.845 unseen under ImageNet inputs and 0.925 under their own min-max (+0.080). Second, with both of its contracts honored, vanilla MedSAM is the strongest entry in the oracle tier. The negative gaps show that this tier measures split difficulty given a location hint, not generalization: a genuinely generalizing model should not score higher on unseen data than seen data. This is the prompt dependence in numbers: the untrained box-prompted model beats its own trained prompt-free version by +0.099 for SAM-ViT-H (0.806 → 0.905) and by +0.200 for corrected MedSAM (0.725 → 0.925). Two conditions change between those numbers, the box arrives and the LoRA training is absent, so each lift reads as the value of location information net of what training added. MedSAM's fine-tuning always supplied a box. The study measures the lift, not the mechanism, but the size of the lift is consistent with an encoder that relies on the box for localization and pays when the box is taken away.

7. Discussion

The adaptation trade is real and favorable off-distribution, on this benchmark family. The result is consistent with the frozen SAM backbone holding visual priors that an ImageNet-initialized ResNet-34 does not pick up from 1,450 polyp images, and with rank-4 LoRA being enough to point them at the task; backbone scale and architecture are not separated here. The result pattern (specialist wins at home, adapter wins away) matches the shape of the cross-dataset question on the one benchmark family I measured, and the drop metric puts the trade in one line: 3% of the trainable parameters comes with about half the generalization drop.

MedSAM is specialized, not weak. The finding I learned the most from is negative in form. MedSAM's published claim (above its specialist baselines on endoscopy) and my measured result (last among the prompt-free models at the time) are both correct. They measure different contracts. MedSAM with a box and min-max inputs posts the highest unseen number in the oracle tier (0.925). MedSAM without the box loses the most (+0.200 from the oracle box), and that size is consistent with an encoder that learned to rely on the prompt for localization; I measured the lift, not the mechanism. A practitioner choosing a model should ask which contract the deployment honors. If no localizer supplies a box at inference, I would pick prompt-free adaptation of generic SAM over medical-pretrained MedSAM at the same size; the 0.036 residual in Section 6.3 is the reason, with the caveat that it rests on one benchmark family and three seeds.

Methods lesson: preprocessing contracts are results. A one-line normalization difference produced a 0.069 Dice artifact that I first read as a scientific finding about medical pretraining. The correction needed a controlled experiment, because the naive fix was itself confounded through the interaction between min-max normalization and photometric augmentation. The pattern I would reuse is to bind preprocessing to the model key in code, keep the published arm, and give the corrected arms separate keys.

Relation to prior work. The result builds on SAMed's approach (LoRA on a frozen SAM encoder for medical segmentation) and adds the measurement SAMed did not make: cross-dataset generalization under the specialists' own protocol. Two parts of the recipe differ here, the CNN decoder in place of SAM's mask decoder and LoRA on the merged qkv projection instead of Q and V separately, so the comparison is to SAMed's approach rather than a replication of it. It also gives PraNet's seen/unseen framing a new use, separating model classes rather than ranking specialist variants.

8. Conclusion

A frozen SAM ViT-H backbone with an 830K-parameter LoRA adapter scores higher than a 24.4M-parameter ImageNet-initialized U-Net on the three unseen PraNet datasets (0.806 vs 0.755 mean Dice, three seeds, no significance test) and cuts the seen-to-unseen drop to about half (0.082 vs 0.145), while the specialist keeps the edge on seen data and in training cost. Controlled ablations split MedSAM's 0.100 unseen deficit against SAM-ViT-B into a 0.069 preprocessing artifact in my own pipeline and a 0.036 residual for the checkpoint at matched

size; a separate 0.045 backbone-size effect, measured on generic SAM, is comparable to that residual. The oracle runs quantify prompt dependence (+0.200 from a ground-truth box for corrected MedSAM against +0.099 for generic SAM).

Contributions. (1) A controlled comparison of three model families (specialist CNN, LoRA-adapted generic SAM at two sizes, LoRA-adapted MedSAM), six trained variants, under one protocol, three seeds, one GPU, with full cost accounting. (2) Two controlled pairings that separate the pretraining source from the backbone size, plus a measured preprocessing effect, via matched arms. (3) A documented root-cause investigation and retraction pattern for a preprocessing confound. (4) A reproducible pipeline: config-driven trainer, GPU-free tests, mechanical results aggregation.

Limitations. I want to be direct about what these numbers do not cover. No significance test was run; the seed spreads in Section 6.1 are the only uncertainty measure. All five splits come from PraNet's datasets, so the results do not speak to colonoscopy data under other equipment or protocols. Three seeds show the spread is real, but the sample is small. The oracle tier has one run per baseline and remains an upper bound, not a peer. Training cost is asymmetric in a way that matters to some deployments (U-Net: 10 min, 98 MB; SAM-H: 120 min, 2.5 GB).

Future work. Four directions, in the order I would take them. The first is already designed and scaffolded; the other three are open.

1. *An ensemble decision system.* The per-split table in Section 6.2 shows different winners on different splits (U-Net on ClinicDB, SAM-ViT-H on every unseen split, and box-prompted MedSAM above both in the oracle tier), which suggests the models make some different errors; the ensemble run will measure how much. The next experiment combines them at inference without reading any ground truth. It has two parts. A **flat ensemble** averages the probability maps of several trained members before the 0.5 threshold; every model already scores on the same 352×352 grid, so the maps average directly. A **box cascade** uses U-Net's prompt-free mask to derive a bounding box, then hands that box to vanilla MedSAM under min-max normalization. This tests whether the 0.925 oracle ceiling in Section 6.4 is reachable with a learned localizer instead of ground truth.

I wrote the design down before the run in `docs/PLAN_ENSEMBLE.md` (on the ensemble branch), with fixed member sets and expected ranges, so any change after the result is visible in the file history. Two member sets are fixed in advance: set A is U-Net plus SAM-ViT-H, the largest architectural distance in the study; set B is the top three prompt-free rows (SAM-ViT-H, SAM-ViT-B, U-Net). Each set runs with uniform weights and with weights fitted on a grid. Weight fitting uses only the first contiguous half of each seen test split, in sorted path order, and holds out the second half. The split is contiguous rather than interleaved because CVC-ClinicDB frames come from video sequences, and an interleaved split would put near-duplicate frames on both sides. The unseen splits stay untouched as the primary endpoint. The decision threshold stays at 0.5, since a fitted threshold would be a second confound. Ensembles and the cascade report in a

third tier, derived, separate from both the trained models and the oracle baselines: they are fair, because they read no ground truth at inference, but they are not peers of a single architecture in a table that asks which backbone generalizes better.

The expected ranges are written down before the run, so a bug is visible. A uniform ensemble should land between 0.77 and 0.83 unseen mDice; a value below 0.75 signals a defect, and a value below SAM-ViT-H's 0.806 should not count as a defect on its own, because I expect U-Net to saturate its sigmoid harder than the LoRA models and pull a uniform average toward itself, which a planned calibration diagnostic will check. Fitted weights should move the uniform result by -0.005 to $+0.015$. The members share training data and loss, so their errors correlate, and I expect a gain of about 0.01 to 0.03 over the best single model rather than a large one. The cascade should land between 0.75 and 0.88: MedSAM with a correct box scores about 0.92, so a detector that produces a usable box on a fraction f of images gives roughly $0.92 f$. Detection caps the cascade, and that cap is the number I most want to measure. Cost is small on the compute side: about one A100-hour for the whole experiment, most of it a 45 to 75 minute Colab session to cache the probability maps for nine runs (the caching compute itself is about 16 minutes), and the rest runs on a CPU from the cache. The cache, combination, and cascade code with its GPU-free tests already exists on a feature branch and is waiting on the GPU pass.

2. *A second benchmark family.* Evaluate outside PraNet's five datasets, since that is the limitation that bounds the claim most.

3. *Color sensitivity.* Probe it with the grayscale-replication experiment already designed in `docs/MEDSAM_INVESTIGATION.md`.

4. *Adapter capacity.* Test larger LoRA ranks and separate Q and V targets, which would also close the remaining difference from the SAMed recipe.

9. References

- [1] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. MICCAI*, 2015, pp. 234–241.
- [2] D.-P. Fan, G.-P. Ji, T. Zhou, G. Chen, H. Fu, J. Shen, and L. Shao, "PraNet: Parallel reverse attention network for polyp segmentation," in *Proc. MICCAI*, 2020, pp. 263–273.
- [3] B. Dong, W. Wang, D.-P. Fan, J. Li, H. Fu, and L. Shao, "Polyp-PVT: Polyp segmentation with pyramid vision transformers," *CAAI Artificial Intelligence Research*, vol. 2, Art. no. 9150015, 2023.
- [4] A. Kirillov et al., "Segment Anything," in *Proc. IEEE/CVF ICCV*, 2023, pp. 4015–4026.
- [5] J. Ma, Y. He, F. Li, L. Han, C. You, and B. Wang, "Segment anything in medical images," *Nature Communications*, vol. 15, Art. no. 654, 2024.

- [6] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," in *Proc. ICLR*, 2022.
- [7] K. Zhang and D. Liu, "Customized Segment Anything Model for medical image segmentation," arXiv:2304.13785, 2023.
- [8] D. Jha, P. H. Smedsrud, M. A. Riegler, P. Halvorsen, T. de Lange, D. Johansen, and H. D. Johansen, "Kvasir-SEG: A segmented polyp dataset," in *Proc. MMM*, 2020, pp. 451–462.
- [9] J. Bernal, F. J. Sánchez, G. Fernández-Esparrach, D. Gil, C. Rodríguez de Miguel, and F. Vilariño, "WM-DOVA maps for accurate polyp highlighting in colonoscopy: Validation vs. saliency maps from physicians," *Computerized Medical Imaging and Graphics*, vol. 43, pp. 99–111, 2015. (CVC-ClinicDB)
- [10] J. Bernal, J. Sánchez, and F. Vilariño, "Towards automatic polyp detection with a polyp appearance model," *Pattern Recognition*, vol. 45, no. 9, pp. 3166–3182, 2012. (CVC-ColonDB)
- [11] J. Silva, A. Histace, O. Romain, X. Dray, and B. Granado, "Toward embedded detection of polyps in WCE images for early diagnosis of colorectal cancer," *Int. J. Comput. Assist. Radiol. Surg.*, vol. 9, no. 2, pp. 283–293, 2014. (ETIS-LaribPolypDB)
- [12] D. Vázquez et al., "A benchmark for endoluminal scene segmentation of colonoscopy images," *Journal of Healthcare Engineering*, vol. 2017, Art. no. 4037190, 2017. (CVC-300 / EndoScene)
- [13] D.-P. Fan, M.-M. Cheng, Y. Liu, T. Li, and A. Borji, "Structure-measure: A new way to evaluate foreground maps," in *Proc. IEEE ICCV*, 2017, pp. 4548–4557.
- [14] D.-P. Fan, C. Gong, Y. Cao, B. Ren, M.-M. Cheng, and A. Borji, "Enhanced-alignment measure for binary foreground map evaluation," in *Proc. IJCAI*, 2018, pp. 698–704.
- [15] D.-P. Fan et al., "PraNet," GitHub repository, 2020. <https://github.com/DengPingFan/PraNet> (training and test split release: 900 Kvasir-SEG and 550 CVC-ClinicDB training images; test splits of 100, 62, 380, 196, and 60 images).
- [16] J. Ma et al., "MedSAM," GitHub repository, 2024. <https://github.com/bowang-lab/MedSAM> (MedSAM_Inference.py: per-image min-max normalization to [0,1]).
- [17] R. Margolin, L. Zelnik-Manor, and A. Tal, "How to evaluate foreground maps?" in *Proc. IEEE CVPR*, 2014, pp. 248–255.
- [18] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. ICLR*, 2019.
- [19] P. Iakubovskii, "Segmentation Models Pytorch," GitHub repository, 2019. https://github.com/qubvel/segmentation_models_pytorch

[20] A. Buslaev, V. I. Iglovikov, E. Khvedchenya, A. Parinov, M. Druzhinin, and A. A. Kalinin, "Albumentations: Fast and flexible image augmentations," *Information*, vol. 11, no. 2, Art. no. 125, 2020.

10. Appendices

Appendix A: Hyperparameters

Setting	Value
Input size	352 × 352
Batch size	4
Optimizer	AdamW, lr 5e-5, weight decay 1e-4
Schedule	cosine
Loss	Dice + BCE
Epoch budget	50, early stop patience 8
Checkpoint selection	best validation Dice
LoRA	r = 4, alpha = 8, dropout 0.1, merged qkv projection of each encoder attention block
Seeds	42, 43, 44
Hardware	NVIDIA A100-SXM4-40GB (Google Colab)
Oracle prompt	ground-truth bounding box, 5 px padding
Normalization	ImageNet stats (SAM, U-Net); per-image min-max [0,1] (corrected MedSAM arms)

Appendix B: Repository guide

Source code: GitHub repository (this project). Entry points: `train.py` (training), `evaluate.py` (5-split evaluation), `zeroshot_eval.py` (oracle-box baselines), `aggregate_results.py` (results consolidation). Full per-run and per-seed numbers: `results/summary/SUMMARY.md`, `summary_flat.csv`, `summary_by_model.csv`. Supporting documents: `docs/FINDINGS.md` (findings write-up), `docs/MEDSAM_INVESTIGATION.md` (hypothesis-driven root-cause analysis), `docs/DECISIONS.md` (design decision log), `docs/PRESENTATION_OUTLINE.md` (talk plan). Tests: `pytest tests/` (70 GPU-free tests, plus 8 transform tests that run only where Albumentations is installed).

Appendix C: Per-run results

The complete per-run table (each model × seed, all five splits, parameters, checkpoint size, epochs, minutes, device) is generated by `aggregate_results.py` and lives at `results/summary/SUMMARY.md`. Qualitative mask overlays for every run live under `results/<model>/seed<seed>/`.